

Complements of Operational Research

6.2 Queueing: limited capacity and several servers

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When space runs out: M/M/1/K

At the station of 6.1 there is room for only **4 cars**: 1 charging and 3 waiting. Anyone who arrives when the 4 places are taken drives away. Now $\rho = \lambda/\mu$ may even be ≥ 1 : the queue never grows without limit.

M/M/1/K ($\rho \neq 1$)

$$P_n = \frac{(1 - \rho) \rho^n}{1 - \rho^{K+1}}, \quad n = 0, \dots, K$$

$$\lambda_{\text{ef}} = \lambda (1 - P_K), \quad W = \frac{L}{\lambda_{\text{ef}}}, \quad W_q = \frac{L_q}{\lambda_{\text{ef}}}$$

$L = \sum n P_n$, $L_q = \sum_{n \geq 1} (n - 1) P_n$. Little's law uses the rate of those who *enter*, λ_{ef} .

K	P_K	λ_{ef} (/h)	L	W (min)
2	0.243	2.27	0.81	21
3	0.154	2.54	1.15	27
4	0.104	2.69	1.44	32
5	0.072	2.78	1.70	37
6	0.051	2.85	1.92	41
∞	0	3	3	60

With $K = 4$, **10.4% of drivers give up** (0.31 per hour); those who stay spend, on average, 32 min. Simulation gives 10.2% and 32 min.

Less space $\Rightarrow$ short waits, but lost customers.

Demand doubles: several servers, M/M/s

With electric cars on the rise, $\lambda = 6$ cars arrive per hour. With one charger ($\mu = 4$), $\rho = 1.5$: the queue grows without end. How many chargers should be installed?

M/M/s ($a = \lambda/\mu$, $\rho = a/s < 1$)

$$P_0 = \left[\sum_{n=0}^{s-1} \frac{a^n}{n!} + \frac{a^s}{s! (1 - \rho)} \right]^{-1}$$

$$P(\text{wait}) = \frac{a^s}{s! (1 - \rho)} P_0, \quad L_q = P(\text{wait}) \frac{\rho}{1 - \rho}$$

$$W_q = L_q / \lambda, \quad W = W_q + 1/\mu, \quad L = \lambda W.$$

s	$P(\text{wait})$	L_q	W_q (min)
2	0.643	1.93	19.3
3	0.237	0.24	2.4
4	0.075	0.045	0.45
5	0.020	0.009	0.09

$P(\text{wait})$ is the **Erlang C formula**, the same one used to size call centres.

Going from 2 to 3 chargers divides the waiting time by 8.

A single queue or one queue per server?

Two chargers, $\lambda = 6/h$ in total:

- **Two separate queues** (each car chooses a charger and stays in that queue): two M/M/1 with $\lambda = 3 \Rightarrow W_q = 45 \text{ min.}$
- **A single queue** (the first in the queue goes to whichever charger becomes free): one M/M/2 with $\lambda = 6 \Rightarrow W_q = 19.3 \text{ min.}$

Same equipment, same demand: the single queue cuts the waiting time to less than half.

Why?

With separate queues one charger may be idle while cars wait at the other. With a single queue this never happens: capacity is shared.

That is why banks, post offices (CTT) and airports use a single queue with a ticket or a serpentine line.

How many chargers? Cost versus waiting

Each charger costs 12 €/h (depreciation, maintenance, contracted power). Each waiting car costs 20 €/h (fleet drivers' time, lost customers).

$$\text{cost}(s) = 12s + 20L_q(s) \quad (\text{€ per hour})$$

s	chargers	waiting	total
2	24	38.57	62.57
3	36	4.74	40.74
4	48	0.90	48.90
5	60	0.17	60.17

Decisions

- **Minimum cost:** 3 chargers (40.74 €/h); 23.7% of cars wait.
- **Service level** “fewer than 10% wait”: 4 chargers (7.5%), at 48.90 €/h.

The difference (8.16 €/h) is the **price of the service level**. The decision belongs to management — the model says how much each option costs.

In the class exercises the waiting cost applies to L_q (those in the queue); if it applied to L , the service time would be added.

```
from math import factorial

def mms(lam, mu, s):
    a = lam / mu; rho = a / s
    P0 = 1 / (sum(a**n / factorial(n) for n in range(s))
              + a**s / (factorial(s) * (1 - rho)))
    Pw = a**s / (factorial(s) * (1 - rho)) * P0    # Erlang C
    Lq = Pw * rho / (1 - rho)
    return Pw, Lq, Lq / lam                        # P(wait), Lq, Wq

lam, mu = 6, 4
for s in range(2, 6):
    Pw, Lq, Wq = mms(lam, mu, s)
    print(s, round(Pw, 3), round(12*s + 20*Lq, 2))
# 2 0.643 62.57 | 3 0.237 40.74 | 4 0.075 48.9 | 5 0.02 60.17
```

With `simula_mms(6, 4, 3)` (chapter file) the simulation gives $W_q \approx 2.4$ min and 23–24% of cars waiting.

The same code solves the supermarket exercise from class (there is a MATLAB and Python version on Moodle).

Example 2 — restaurant (M/M/1/GD/K/∞)

At the peak hour, a restaurant receives, on average, 400 customers per hour and has a mean service rate, within its quality standards, of 600 customers per hour. Solve with the M/M/1/GD/K/∞ model with limit $K = 300$; exponential arrivals and service.

- a) What is the mean queue length?
- b) What is the probability that there is a queue?
- c) What is the mean time a customer spends in the system?
- d) Compare with the M/M/1 model (no limit). Does the limit $K = 300$ make a difference? Why?

Exercise sheet 6 — Exercise 1 (homework)

How many checkouts should a supermarket open? (homework)

A supermarket needs to decide how many checkouts should be open. On average, 18 customers arrive per hour and a customer is served, on average, in 4 minutes; exponential arrivals and service ($M/M/s/GD/\infty/\infty$). The hourly wage cost is 20 m.u. and the cost per waiting customer is 0.25 m.u. per minute.

- a) How many checkouts should be open to minimise the total cost?
- b) How many checkouts ensure that the probability of a customer having to wait is less than 5%?
What is the total cost in that configuration?

Submission: by email to the lecturer, by the deadline given in class. All exercises: **Exercise sheet 6**.

KEY TAKEAWAYS — 6.2

1. M/M/1/K: $P_n \propto \rho^n$ up to K ; anyone arriving when the system is full is lost; Little with $\lambda_{\text{ef}} = \lambda(1 - P_K)$.
2. M/M/s: P_0 , $P(\text{wait})$ (Erlang C), $L_q = P(\text{wait}) \rho / (1 - \rho)$; requires $\lambda < s\mu$.
3. A single queue is better than separate queues with the same servers.
4. Sizing: minimise server cost + waiting cost, or meet a service level — and know how much the difference costs.

Next: 6.3 When customers are few: finite population (M/M/R)

References Full list: *Course bibliography*.

- Hillier, F. S., Lieberman, G. J., *Introduction to Operations Research*, 11th ed., McGraw-Hill, 2021 — Queueing Theory (M/M/s, finite capacity, decision models).
- Gross, D., Shortle, J. F., Thompson, J. M., Harris, C. M., *Fundamentals of Queueing Theory*, 4th ed., Wiley, 2008.
- Taha, H. A., *Operations Research: An Introduction*, 10th ed., Pearson, 2017 — cost models.

Resources

- Exercise sheet 6. `code/python/06_queueing/06_2_mmk_mms/filas_mms.py`.